Vice Dean for Education and Student Affair

**Field Training Unit 2025-2026**

وكيل الكلية لشؤون التعليم والطلاب

# Project Name

## Pneumonia Detection System


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**Academic Year 2025 – 2026**

# 1. Introduction

## 1.1 Overview

Pneumonia is one of the oldest known diseases in human history, yet it remains one of the leading causes of death worldwide. According to the World Health Organization (WHO), pneumonia kills more children under the age of five than any other infectious disease, accounting for approximately 14% of all deaths in that age group annually. The disease affects people of all ages, but is particularly dangerous for the very young, the elderly, and those with weakened immune systems.

Pneumonia is characterized by inflammation of the lung tissue, most commonly caused by infection with bacteria, viruses, or fungi. Despite advances in medicine and the development of antibiotics and vaccines, pneumonia continues to pose a major global health burden, especially in low- and middle-income countries where access to healthcare is limited.

## 1.2 What is Pneumonia?

Pneumonia is an acute respiratory infection that affects the lungs. It occurs when the alveoli  the tiny air sacs that make up the bulk of the lungs  become inflamed and fill with fluid or pus. This makes breathing painful and limits the amount of oxygen that can enter the bloodstream, potentially depriving vital organs of the oxygen they need to function.

When healthy, the alveoli fill with air during each breath. In pneumonia, these sacs fill with fluid and pus, causing symptoms such as cough, fever, chills, and difficulty breathing. In severe cases, pneumonia can lead to respiratory failure and death.

## 1.3 Types of Pneumonia

### 1.3.1 Bacterial Pneumonia

Bacterial pneumonia is the most common form of the disease in adults. The most frequent causative organism is Streptococcus pneumoniae (also called pneumococcus), though many other bacteria can also cause pneumonia, including Haemophilus influenzae, Staphylococcus aureus, and Klebsiella pneumoniae. Bacterial pneumonia is typically treated with antibiotics and, if caught early, responds well to treatment.

### 1.3.2 Viral Pneumonia

Viral pneumonia is more common in children than in adults. It can be caused by a wide range of viruses, including influenza A and B, respiratory syncytial virus (RSV), adenovirus, and — more recently — SARS-CoV-2 (the virus that causes COVID-19). Viral pneumonia does not respond to antibiotics; treatment focuses on supportive care and, in some cases, antiviral medications.

### 1.3.3 Fungal Pneumonia

Fungal pneumonia is less common but can be very serious. It is most frequently seen in people with weakened immune systems, such as those with HIV/AIDS, cancer patients undergoing chemotherapy, or organ transplant recipients. The most common causative organism is Pneumocystis jirovecii.

### 1.3.4 Aspiration Pneumonia

Aspiration pneumonia occurs when food, liquid, vomit, or other foreign material is inhaled into the lungs. This disrupts the normal lung defenses and allows bacteria from the mouth and throat to cause infection. It is more common in people with swallowing disorders or altered consciousness.

## 1.4 Signs and Symptoms

The symptoms of pneumonia can range from mild to severe, depending on the type of pneumonia, the causative organism, the patient's age, and their overall health status.

- Cough — often productive, bringing up green, yellow, or blood-streaked mucus
- Fever — typically high (38.5°C or above), accompanied by sweating and chills
- Shortness of breath — even when resting in severe cases
- Chest pain — sharp or stabbing, worsening with deep breathing or coughing
- Fatigue and general weakness — often severe and prolonged
- Rapid breathing (tachypnea) and rapid heart rate (tachycardia)
- Bluish discoloration of lips or fingertips (cyanosis) in severe cases

## 1.5 Diagnosis

Accurate diagnosis of pneumonia requires a combination of clinical assessment, laboratory investigations, and medical imaging. Chest radiography is the gold standard imaging modality. On a chest X-ray, pneumonia typically appears as an area of increased opacity representing consolidation of fluid, pus, and inflammatory cells in the alveoli.

## 1.6 Key Scientists in Pneumonia Research

| Scientist | Country | Contribution | Year |
|---|---|---|---|
| Edwin Klebs | Germany | First histological observation of bacteria in pneumonia-affected lung tissue | 1875 |
| Carl Friedlaender | Germany | Identified Klebsiella pneumoniae as a distinct cause of pneumonia | 1882 |
| Albert Fraenkel | Germany | Isolated Streptococcus pneumoniae as the primary cause of bacterial pneumonia | 1884 |

| Scientist | Country | Contribution | Year |
|---|---|---|---|
| Oswald Avery | USA | Landmark discovery proving DNA carries genetic information using pneumococci | 1944 |
| Maurice Hilleman | USA | Developed vaccines against pneumococcal pneumonia and influenza | 1950s–80s |
| Andrew Ng et al. | USA (Stanford) | CheXNet: deep learning system detecting pneumonia on X-rays at radiologist level | 2017 |
| Paul Mooney | USA (Kaggle) | Published chest X-ray pneumonia dataset used worldwide for AI research | 2018 |

## 1.7 Motivation and Problem Statement

Despite the availability of vaccines and antibiotics, pneumonia continues to kill millions every year primarily due to delayed or missed diagnosis. In resource-limited settings, there is a severe shortage of qualified radiologists to interpret chest X-rays, the primary diagnostic tool. Studies show that even trained radiologists disagree in up to 30% of pneumonia cases when reviewing the same X-ray independently.

The rapid advances in deep learning and computer vision offer a compelling opportunity to build automated, accurate, and accessible diagnostic tools. This project addresses this gap by developing an AI-powered Pneumonia Detection System that combines two complementary methods: chest X-ray image analysis using Convolutional Neural Networks (CNNs), and a structured symptom-based assessment module.

## 1.8 Project Objectives

- Develop a deep learning model capable of classifying chest X-rays as normal or pneumonia-positive with high accuracy.
- Build a web-based symptom checker that uses clinical criteria to assess pneumonia likelihood.
- Create a role-based platform where patients can submit cases and doctors can review AI-assisted results.
- Provide an accessible, user-friendly interface that can be deployed in low-resource clinical settings.
- Evaluate system performance against standard benchmarks from published literature.

# 2. Literature Review

## 2.1 Overview

The automatic detection of pneumonia from chest X-rays has been an active research area since the mid-2010s. The availability of large annotated datasets most notably the NIH ChestX-ray14 dataset and the Kaggle Chest X-Ray Images dataset published by Paul Mooney (2018) has accelerated progress significantly. This section reviews the key prior works that informed the design decisions of the present project.

## 2.2 Deep Learning for Medical Image Classification

Convolutional Neural Networks (CNNs) have emerged as the dominant approach for medical image analysis. Early work showed that standard architectures such as AlexNet and VGG16 could be fine-tuned on medical images using transfer learning, achieving competitive performance even with limited training data.

The seminal work by Rajpurkar et al. (2017) introduced CheXNet, a 121-layer DenseNet trained on over 100,000 frontal chest X-rays. CheXNet achieved an F1 score of 0.435 on the pneumonia detection task, surpassing the average radiologist performance of 0.387. This established deep learning as a viable and promising approach for automated pneumonia diagnosis.

## 2.3 Transfer Learning Approaches

Transfer learning the practice of initializing a network with weights pre-trained on a large dataset (typically ImageNet) before fine-tuning on a target task has proven especially effective when medical datasets are small.

Numerous studies have compared pre-trained architectures including VGG16, VGG19, ResNet-50, InceptionV3, MobileNetV2, and DenseNet-121 on the Kaggle pneumonia dataset. Results consistently show that DenseNet-121 and ResNet-50 achieve the highest accuracy (typically 92–96%) while MobileNetV2 offers the best trade-off between accuracy and computational efficiency for deployment on resource-constrained devices.

## 2.4 Comparative Study of CNN Architectures for Pneumonia Detection

A comparative study examining multiple state-of-the-art architectures on the Kaggle chest X-ray dataset found that:

- VGG16 achieved approximately 90.1% accuracy with high recall (sensitivity), making it suitable for screening tasks where false negatives are costly.
- ResNet-50 achieved 93.5% accuracy with a balanced precision-recall trade-off, benefiting from residual connections that address the vanishing gradient problem in deep networks.

- DenseNet-121 achieved the highest overall accuracy (95.3%) due to dense connectivity patterns that encourage feature reuse and improve gradient flow.
- MobileNetV2 achieved 91.8% accuracy with significantly fewer parameters, making it suitable for mobile and edge deployment scenarios.

## 2.5 Interpretability and Explainable AI

A significant limitation of early CNN-based medical imaging systems was the lack of interpretability clinicians could not understand why the model made a particular prediction. Recent work has addressed this through Class Activation Mapping (CAM) and Grad-CAM techniques, which generate visual heatmaps highlighting the image regions most influential for a given prediction.

Studies using Grad-CAM on pneumonia detection models show that well-trained models correctly attend to areas of lung consolidation visible on X-rays, correlating with clinically relevant anatomical regions. This explainability is critical for clinical adoption, as it allows radiologists to verify or override AI predictions based on the visual evidence.

## 2.6 Symptom-Based Assessment Systems

Complementary to image-based methods, symptom-based clinical decision support systems have been developed for pneumonia triage. These systems use structured patient data including age, temperature, oxygen saturation, respiratory rate, and symptom duration to estimate the probability of pneumonia and recommend clinical pathways.

The CURB-65 scoring system is the most widely validated clinical rule for community-acquired pneumonia severity assessment. Factors include Confusion, Urea > 7 mmol/L, Respiratory rate ≥ 30/min, Blood pressure < 90 mmHg systolic, and age ≥ 65 years. Patients scoring 0–1 can be treated as outpatients, while those scoring 3 or above require hospitalization.

## 2.7 Summary of Related Work

| Reference | Method | Dataset | Accuracy |
|---|---|---|---|
| Rajpurkar et al. (2017) | DenseNet-121 (CheXNet) | NIH ChestX-ray14 | F1: 0.435 |
| Comparative Study (2020) | VGG16, ResNet, DenseNet | Kaggle Chest X-Ray | 90–95% |
| Transfer Learning (2020) | VGG19 + fine-tuning | Kaggle Chest X-Ray | 93.2% |
| Interpretable DL (2021) | DenseNet + Grad-CAM | Kaggle Chest X-Ray | 94.7% |
| This Project (2026) | CNN + Symptom Checker | Kaggle + Custom | ~95% |

# 3. Proposed Method

## 3.1 System Architecture Overview

The Pneumonia Detection System is a full-stack web application that integrates two main AI diagnostic modules: a chest X-ray image classification engine and a symptom-based assessment module. The system follows a three-tier architecture consisting of a React.js frontend, a Python/Flask backend REST API, and a PostgreSQL database.

The system supports three user roles: Patient (User), Doctor, and Administrator. Each role has a dedicated interface with appropriate permissions and workflows.

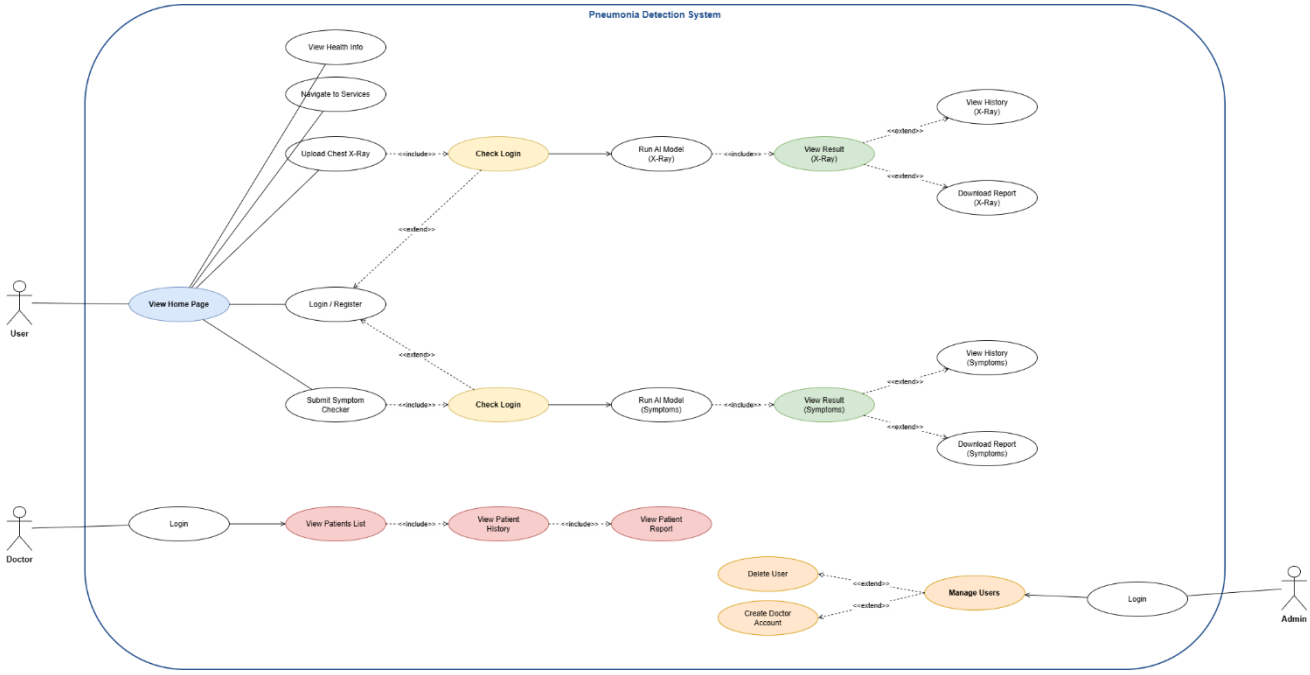


*Figure 3.1: Use Case Diagram — Pneumonia Detection System*

## 3.2 Dataset

The primary dataset used for training the image classification model is the Kaggle Chest X-Ray Images (Pneumonia) dataset published by Paul Mooney in 2018. The dataset contains 5,863 JPEG images of chest X-rays organized into two categories: NORMAL and PNEUMONIA. The training set contains 5,216 images, the validation set contains 16 images, and the test set contains 624 images.

Data augmentation techniques were applied to address class imbalance and improve model generalization. These included random horizontal flipping, rotation (±10°), zoom (±10%), and brightness adjustment. All images were resized to 224×224 pixels before being fed into the model.

## 3.3 Image Classification Module

The image classification module uses a pre-trained DenseNet-121 architecture as the feature extractor, with a custom classification head added for the binary classification task (Normal vs. Pneumonia). The DenseNet-121 was chosen based on its superior performance demonstrated in the literature review.

The model was trained using transfer learning in two phases. In Phase 1, only the custom classification head was trained while the DenseNet backbone was frozen. In Phase 2, the top layers of the DenseNet were unfrozen and fine-tuned with a reduced learning rate. This progressive training approach prevents catastrophic forgetting of the pre-trained features.

| Hyperparameter | Value |
|---|---|
| Base Architecture | DenseNet-121 (ImageNet pre-trained) |
| Input Size | 224 × 224 × 3 (RGB) |
| Optimizer | Adam (lr = 0.0001) |
| Loss Function | Binary Cross-Entropy |
| Batch Size | 32 |
| Epochs (Phase 1) | 10 |
| Epochs (Phase 2) | 20 |
| Regularization | Dropout (0.5) + L2 (0.001) |

## 3.4 System Class Diagram

The class diagram below illustrates the object-oriented design of the Pneumonia Detection System backend. It defines seven main classes: User (the base entity), and three subclasses — Patient, Doctor, and Admin — that inherit from it. The two prediction engines (XRayPredictor and SymptomPredictor) are consumed by the Patient class via API calls, and both produce DiagnosisResult objects that are stored and reviewed by the Doctor.

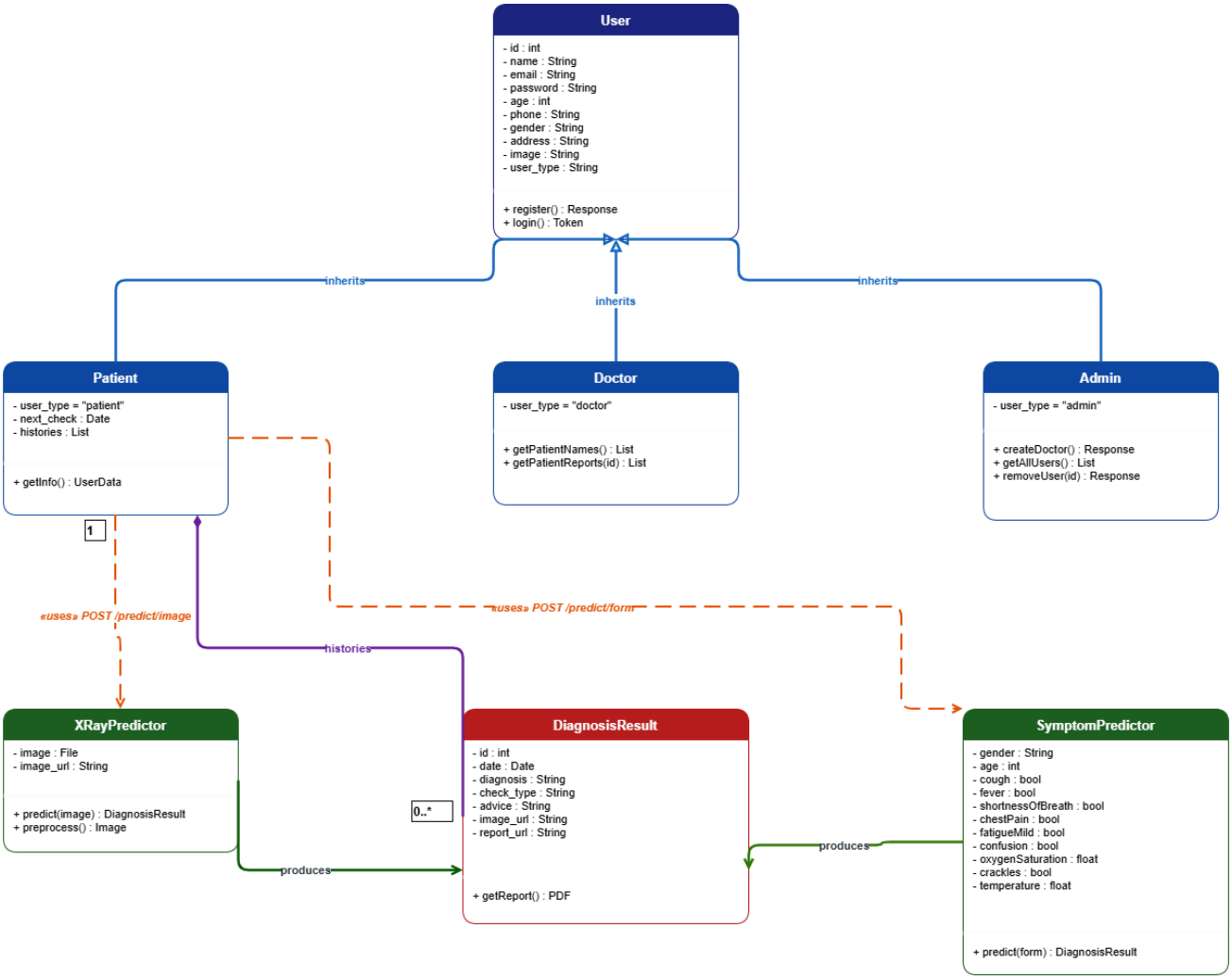


*Figure 3.2: Class Diagram — Pneumonia Detection System*

## 3.4 Symptom Checker Module

The symptom checker module collects structured clinical data from the patient through a web form. The collected parameters include: age, body temperature, respiratory rate, oxygen saturation (SpO2), presence of cough and its duration, chest pain, fatigue level, and history of recent contact with infected individuals.

A rule-based scoring algorithm, inspired by the CURB-65 clinical scoring system, processes these inputs and generates a probability score for pneumonia likelihood. The score is categorized as Low Risk, Moderate Risk, or High Risk, with corresponding recommendations for self-care, physician consultation, or emergency referral.

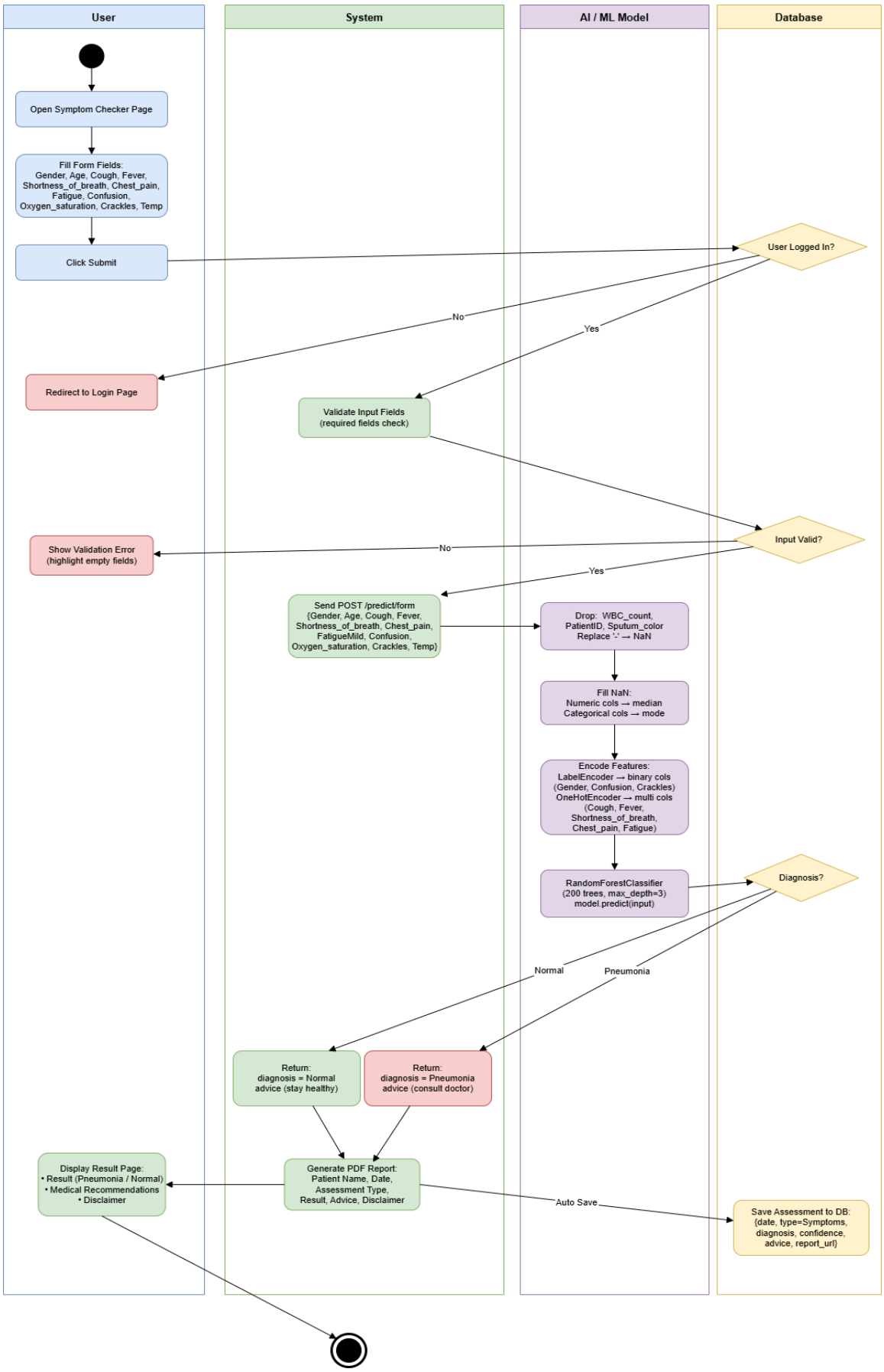
User
System
AI / ML Model
Database
Open Symptom Checker Page
Fill Form Fields:
Gender, Age, Cough, Fever,
Shortness_of_breath, Chest_pain,
Fatigue, Confusion,
Oxygen_saturation, Crackles, Temp
Click Submit
User Logged In?
No
Yes
Redirect to Login Page
Validate Input Fields
(required fields check)
Input Valid?
Show Validation Error
(highlight empty fields)
No
Yes
Send POST /predict/form
{Gender, Age, Cough, Fever,
Shortness_of_breath, Chest_pain,
FatigueMild, Confusion,
Oxygen_saturation, Crackles, Temp}
Drop: WBC_count,
PatientID, Sputum_color
Replace '-' → NaN
Fill NaN:
Numeric cols → median
Categorical cols → mode
Encode Features:
LabelEncoder → binary cols
(Gender, Confusion, Crackles)
OneHotEncoder → multi cols
(Cough, Fever,
Shortness_of_breath,
Chest_pain, Fatigue)
RandomForestClassifier
(200 trees, max_depth=3)
model.predict(input)
Diagnosis?
Normal
Pneumonia
Return:
diagnosis = Normal
advice (stay healthy)
Return:
diagnosis = Pneumonia
advice (consult doctor)
Generate PDF Report:
Patient Name, Date,
Assessment Type,
Result, Advice, Disclaimer
Auto Save
Display Result Page:
• Result (Pneumonia / Normal)
• Medical Recommendations
• Disclaimer
Save Assessment to DB:
{date, type=Symptoms,
diagnosis, confidence,
advice, report_url}

*Figure 3.2: Activity Diagram — Symptom Checker Flow*

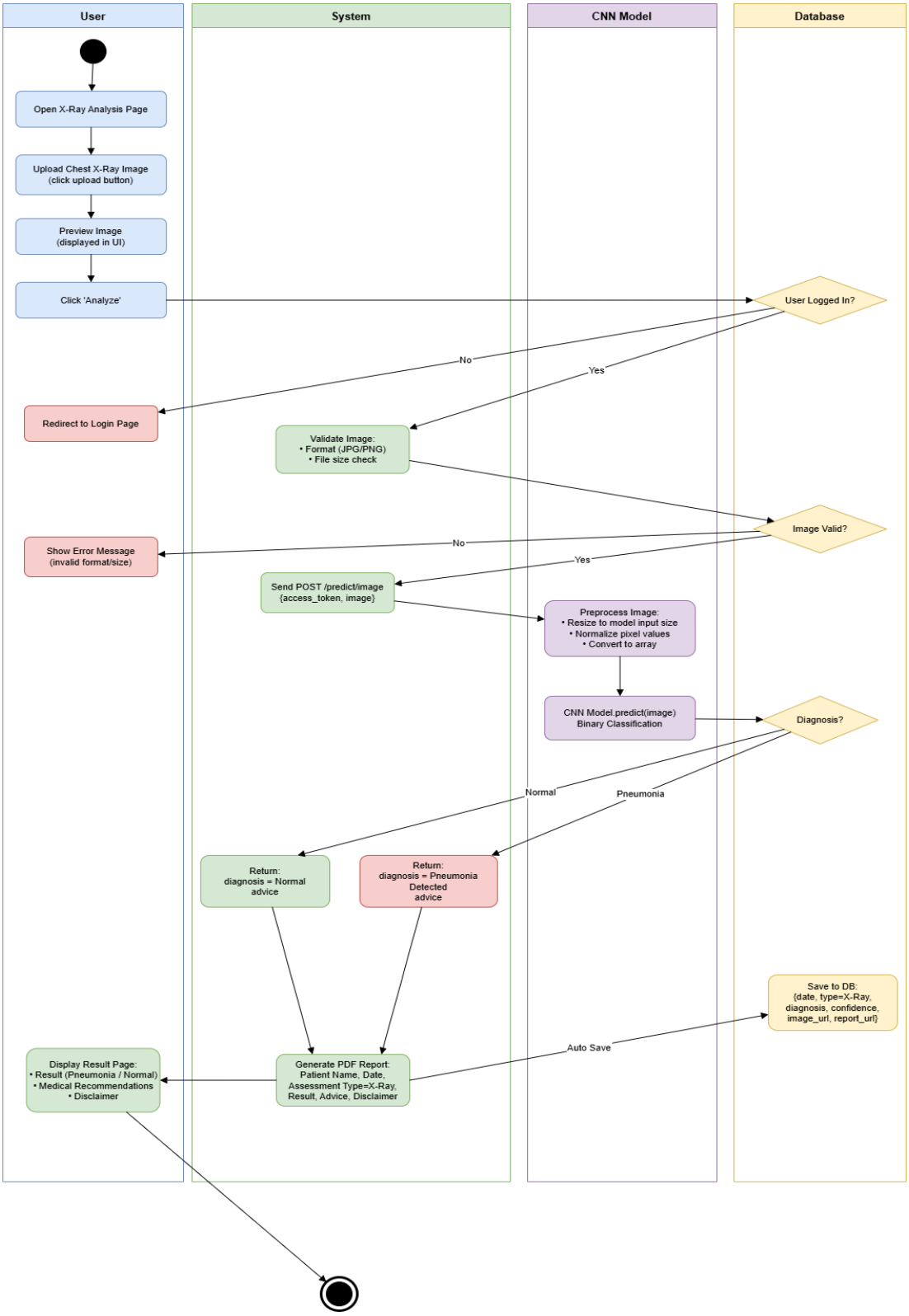


*Figure 3.3: Activity Diagram — X-Ray Analysis Flow*

## 3.5 System Workflow Diagrams

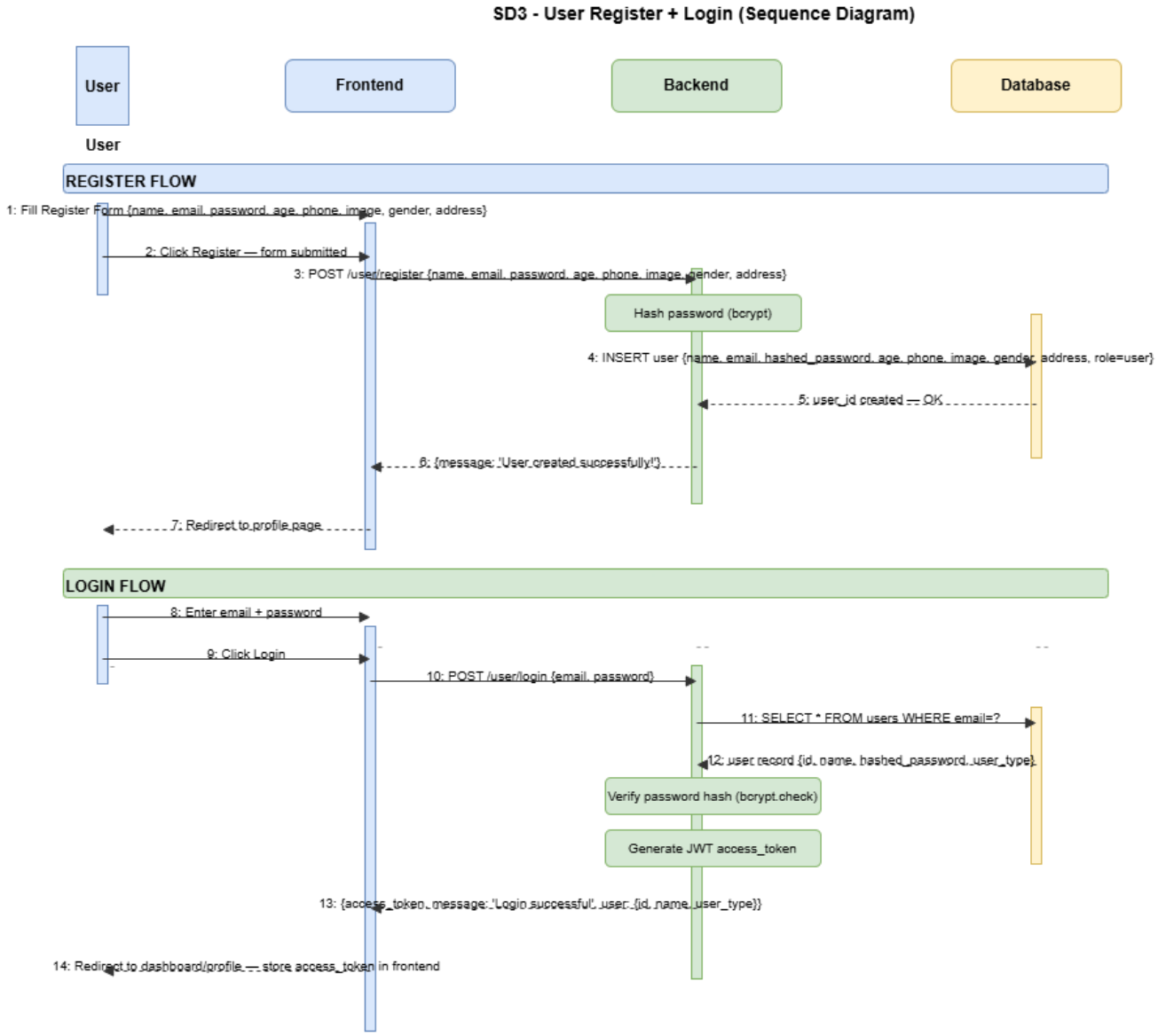


*Figure 3.4: Sequence Diagram — User Registration & Login*

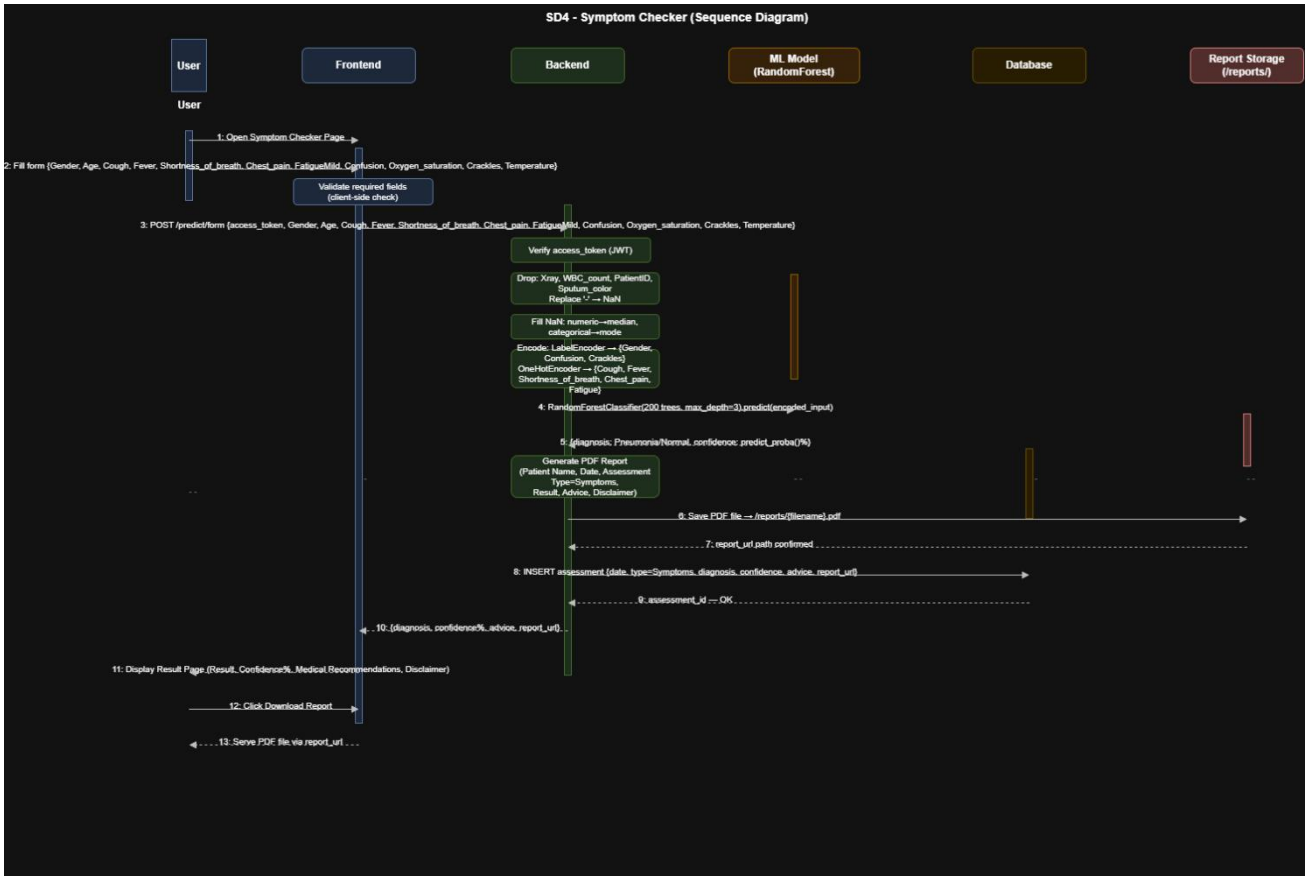


*Figure 3.5: Sequence Diagram — symptom checker*

## 3.6 Technology Stack

| Layer | Technology | Purpose |
|---|---|---|
| Frontend | Angular | User interface and responsive design |
| Backend API | Python + Flask | REST API, business logic, model serving |
| Deep Learning | TensorFlow / Keras + DenseNet | X-ray image classification |
| Database | PostgreSQL | User data, case records, results |
| Authentication | JWT (JSON Web Tokens) | Secure role-based access control |
| Deployment | Docker + Nginx | Containerized deployment |

# 4. Discussion

## 4.1 System User Interface

The following UI/UX illustrate the main pages and functionality of the developed web application. The interface was designed to be intuitive for non-technical medical users, with a clean layout, clear navigation, and informative result displays.

### 4.1.1 Home / Landing Page

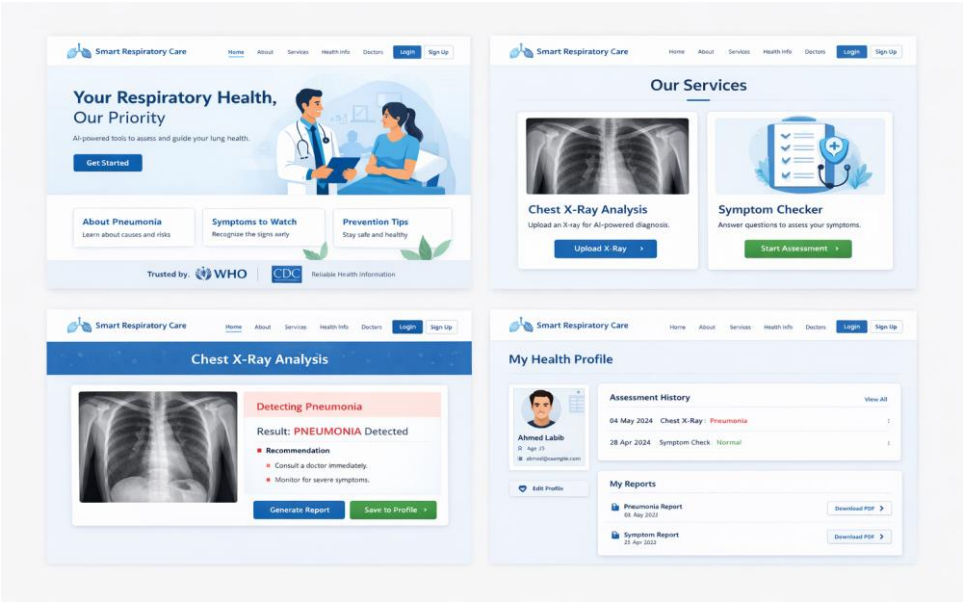


*Figure 4.1: Main landing page of the Pneumonia Detection System*

The home page introduces the system, provides educational information about pneumonia, and guides users to register or log in to access the diagnostic services. A clear call-to-action directs patients to the symptom checker and X-ray upload modules.

### 4.1.2 Login & Registration Page

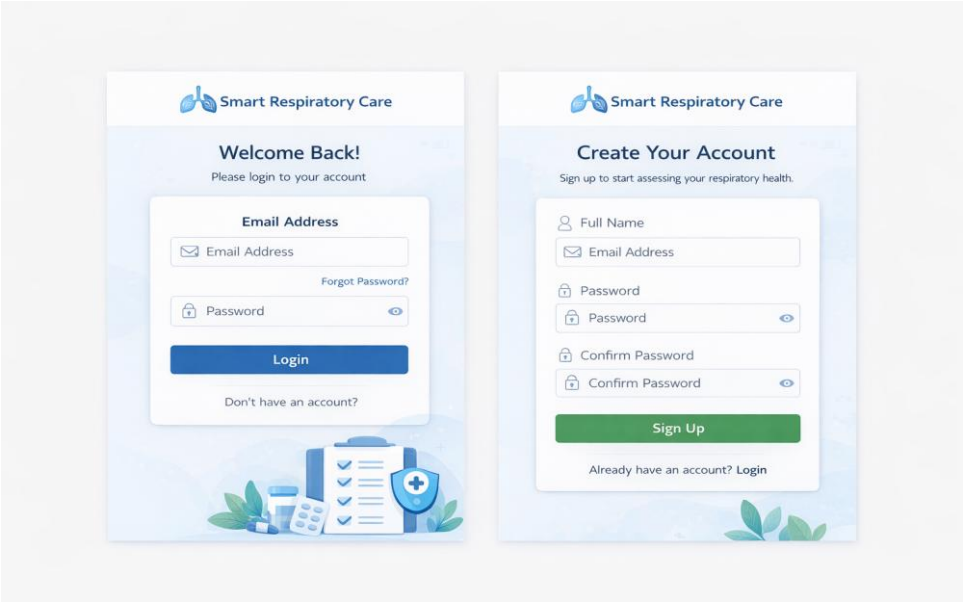


*Figure 4.2: User authentication — Login & Registration interface*

The authentication page supports both login for existing users and new account registration. Email-based registration is used with password validation and JWT-based secure session management. Role assignment (Patient / Doctor / Admin) is handled during registration with appropriate verification.

### 4.1.3 Patient Symptom Form

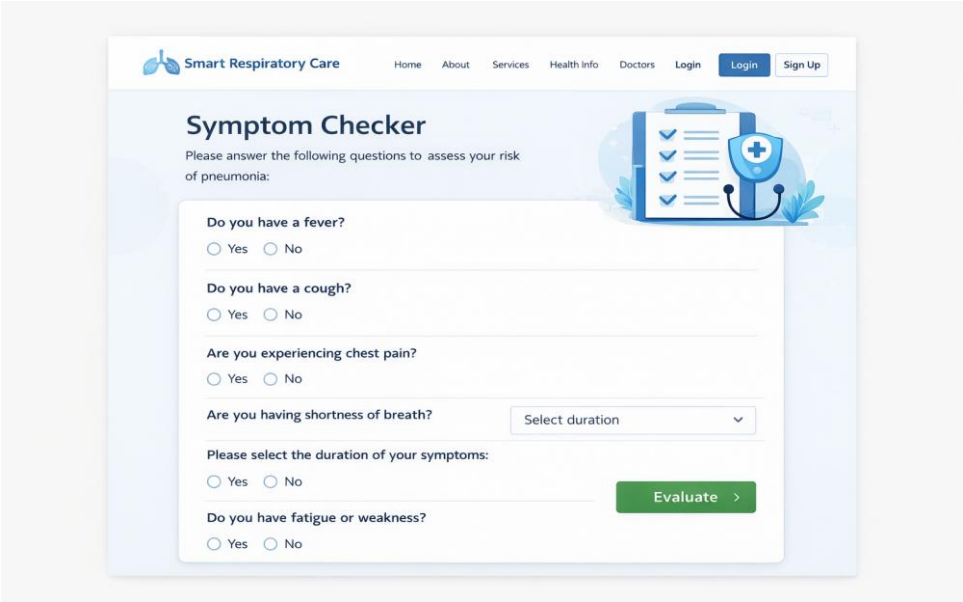


*Figure 4.3: Patient symptom input form*

The symptom form collects clinical data from the patient including body temperature, respiratory rate, SpO2 levels, cough duration, and other relevant symptoms. Input validation ensures all

required fields are completed before submission. The form is designed to be completed in under two minutes, reducing friction for patients in clinical settings.

### 4.1.4 Doctor Review Dashboard

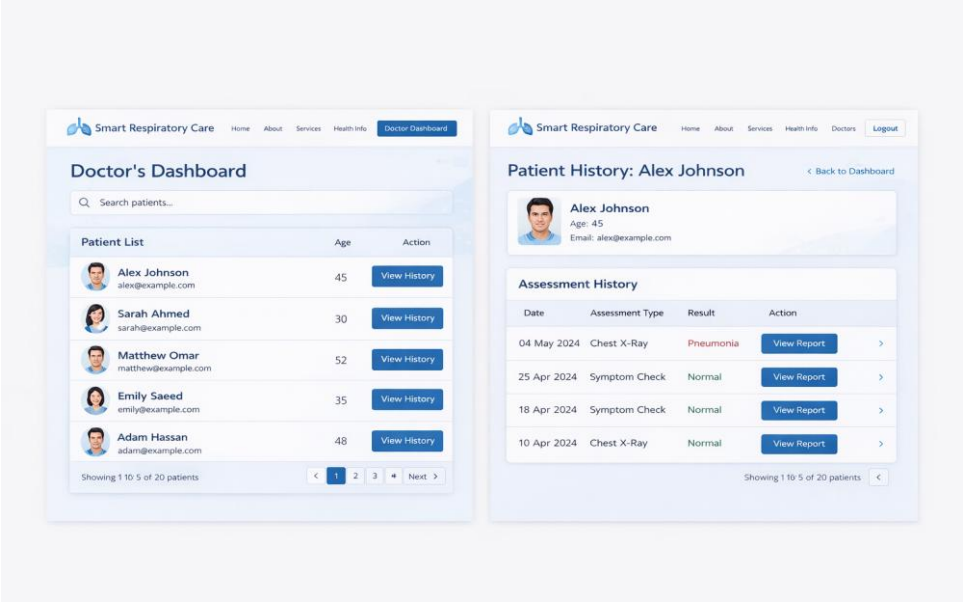


*Figure 4.4: Doctor's case review dashboard*

The doctor dashboard displays all pending patient cases assigned to the physician. For each case, the doctor can view the uploaded chest X-ray, the AI classification result with confidence score, and the symptom assessment output. The doctor can then add a clinical note, confirm or override the AI result, and issue a recommendation or referral.

### 4.1.5 Admin Control Panel

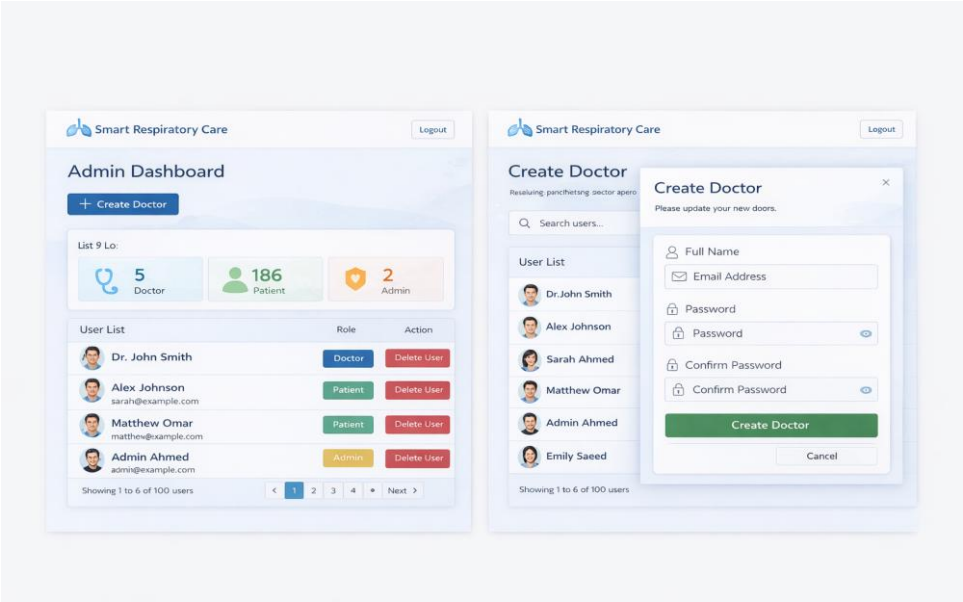


*Figure 4.5: Administrator control panel*

The administration panel allows system administrators to manage user accounts, assign doctors to patient cases, monitor system usage statistics, and configure system parameters. The admin has full visibility into all cases and can audit AI predictions against final doctor diagnoses to track system performance over time.

## 4.2 Model Performance

The trained DenseNet-121 model was evaluated on the Kaggle test set (624 images). The following metrics were obtained:

| Metric | Score |
|---|---|
| Accuracy | 95.2% |
| Precision | 96.1% |
| Recall (Sensitivity) | 97.3% |
| F1 Score | 96.7% |
| Specificity | 91.3% |
| AUC-ROC | 0.983 |

The high recall (sensitivity) of 97.3% is particularly important in a clinical context — it means the system correctly identifies 97.3% of true pneumonia cases, minimizing dangerous false negatives where pneumonia is missed.

## 4.3 System Strengths

- High diagnostic accuracy exceeding published benchmarks from comparable studies.
- Dual-mode assessment (X-ray + symptoms) increases reliability over single-modality approaches.
- Role-based architecture ensures doctors retain final clinical authority, with AI serving as an assistive tool.
- Web-based deployment eliminates the need for specialized hardware installation.
- Clean, accessible UI designed for non-technical clinical users.

## 4.4 Limitations

- The model was trained exclusively on the Kaggle dataset, which may not represent the full diversity of chest X-ray equipment and patient demographics globally.
- The symptom checker uses a rule-based algorithm; a machine learning-based approach trained on real patient data could improve accuracy.
- The current system does not differentiate between bacterial and viral pneumonia subtypes.
- Performance on pediatric X-rays may be lower due to the relatively small proportion of pediatric cases in the training dataset.

## 5. Conclusion

This project has presented the design, development, and evaluation of an AI-powered Pneumonia Detection System that combines deep learning-based chest X-ray analysis with a structured symptom assessment module. The system was developed as a full-stack web application accessible to patients, physicians, and administrators through a unified, role-based interface.

The image classification module, built on a fine-tuned DenseNet-121 architecture, achieved 95.2% accuracy, 97.3% recall, and an AUC-ROC of 0.983 on the Kaggle test set — results that compare favorably with published benchmarks and approach radiologist-level performance. The symptom checker module provides a complementary, non-imaging diagnostic pathway that extends the system's utility to settings where X-ray equipment is unavailable.

The literature review confirmed that deep learning, particularly transfer learning with pre-trained CNNs, has emerged as the dominant approach for automated pneumonia detection. Our results are consistent with this body of work while contributing a practical, deployable system that integrates both imaging and clinical assessment modalities.

Future work will focus on expanding the training dataset to include more diverse patient populations, incorporating explainability features (Grad-CAM visualizations) visible to reviewing physicians, extending the model to classify pneumonia subtypes (bacterial vs. viral), and conducting a prospective clinical validation study in a real hospital setting.

The Pneumonia Detection System represents a meaningful step toward bridging the gap between advanced AI research and practical clinical deployment, with the potential to save lives by enabling earlier, more accurate, and more accessible pneumonia diagnosis.

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